ClipSAT : ACT International Subject Test: Mathematics 2 Level 2

Unit 6.01: Limits, Asymptotes & Instantaneous Rates

Name: _____

Q1. The graph of the function fx = x - 1^2 - 4 is shown in the standard x, y coordinate plane below. What is the range of f ?

[ ] A: [-4, infty
[ ] B: [1, infty
[ ] C: -infty, -4]
[ ] D: -infty, infty

Q2. For the complex number i = sqrt{-1}, which of the following is equivalent to the expression 5 - 3i - 2 - 7i?

[ ] A: 3 + 4i
[ ] B: 3 - 10i
[ ] C: 7 + 4i
[ ] D: 3 - 4i

Q3. What is the value of lim_{x to 3} frac{x^2 - 2x - 3}{x - 3} ?

[ ] A: 4
[ ] B: 0
[ ] C: 6
[ ] D: The limit does not exist

Q4. Given the vectors mathbf{u} = langle 3, -4 rangle and mathbf{v} = langle -2, 5 rangle , what is the vector 2mathbf{u} + mathbf{v} ?

[ ] A: langle 4, -3 rangle
[ ] B: langle 1, 1 rangle
[ ] C: langle 4, 1 rangle
[ ] D: langle 8, -13 rangle

Q5. The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

[ ] A: 131
[ ] B: 138
[ ] C: 140
[ ] D: 145

Q6. A point has polar coordinates r, theta = left6, frac{2pi}{3}right. What are the Cartesian coordinates x, y of this point?

[ ] A: -3, 3sqrt{3}
[ ] B: -3sqrt{3}, 3
[ ] C: 3, -3sqrt{3}
[ ] D: 3sqrt{3}, -3

Q7. The polynomial function Px = 2x^3 + ax^2 - 5x + b has x - 1 as a factor. When Px is divided by x + 2 , the remainder is -18 . What is the remainder when Px is divided by x - 3 ?

[ ] A: 2
[ ] B: 8
[ ] C: 18
[ ] D: -30

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Q8. For what real value of k is the complex expression frac{k + 8i}{3 - i} + frac{2 - ki}{3 + i} a real number?

[ ] A: 11

[ ] B: 1

[ ] C: 6

[ ] D: -11